Table: Developers

DevID	DevName	Department	BaseSalary	JoinDate	Email
1	Alice Smith	Frontend	55000	2021-05-10	alice@tech.com
2	Bob Johnson	Backend	72000	2020-08-15	bob@dev.net
3	Charlie Brown	DevOps	65000	2022-01-20	charlie@tech.com
4	Diana Prince	Backend	80000	2019-11-05	diana@dev.net
5	Evan Wright	Frontend	48000	2023-03-01	evan@tech.com
6	Fiona Gallagher	Data Science	95000	2018-07-22	fiona@data.org
7	George Lucas	Backend	61000	2022-09-10	george@dev.net
8	Hannah Abbott	DevOps	68000	2021-12-12	hannah@tech.com
9	Ian Somerhalder	Frontend	52000	2022-05-30	ian@tech.com
10	Jane Doe	Data Science	89000	2020-02-18	jane@data.org

Table: Project

ProjectID	ProjectName	Client	Budget	Status
P1	Project Alpha	TechCorp	500000	Active
P2	Project Beta	HealthInc	300000	Active
P3	Project Gamma	FinBank	750000	Completed
P4	Project Delta	EduNet	200000	Active

Table: ProjectAllocations

AllocID	DevID	ProjectID	AllocHours	Role
101	2	P1	120	Lead
102	1	P1	80	Developer
103	4	P2	150	Lead
104	6	P3	200	Data Architect
105	3	P1	100	Ops Manager
106	7	P2	90	Developer
107	10	P3	180	Analyst
108	5	P4	60	UI/UX
109	8	P2	110	Ops Manager
110	9	P4	70	Developer

Varendra University

Department of Computer Science and Engineering

5th Semester Mid Lab Test (Spring-2026)

Course Code: CSE 3104

Course Title: Database Management System Lab

Time: 45 minutes

Marks: 24 (3x8)

Question no	CO's no.	Description	Taxonomy domain/level	POs	K
1	CO1	Utilize SQL commands to perform basic database tasks	Cognitive/ Apply	PO1	WK4

1. **Execute** an SQL query to increase the BaseSalary by 15% strictly for developers who belong to the 'Backend' department and joined the company before January 1, 2022, ensuring that this update does not push any developer's salary beyond a maximum ceiling of 85,000.
2. **Execute** a single SQL query utilizing built-in functions to display the developer's name in entirely uppercase letters, the length of their email address, and the exact total number of months they have been employed since their JoinDate up to the current system date, filtering the results to only show developers who have worked for more than 24 months.
3. **Execute** an SQL query to retrieve the DevName, Department, and BaseSalary of the developers from the database, strictly skipping the top 3 highest-paid developers in the company and fetching only the data of the subsequent 4 highest-paid developers.
4. **Execute** an SQL query using aggregate functions to calculate the total number of developers assigned and the total sum of AllocHours dedicated to each ProjectID from the ProjectAllocations table, displaying only those projects where the total allocated hours strictly exceed 150 hours, and ensure the final output is sorted in descending order based on the total allocated hours.
5. **Execute** the necessary SQL statements to alter the structure of the existing Developers table to enforce data integrity by adding a unique constraint to the Email column and a check constraint to ensure that the BaseSalary can never be inserted or updated to a value lower than 45,000.